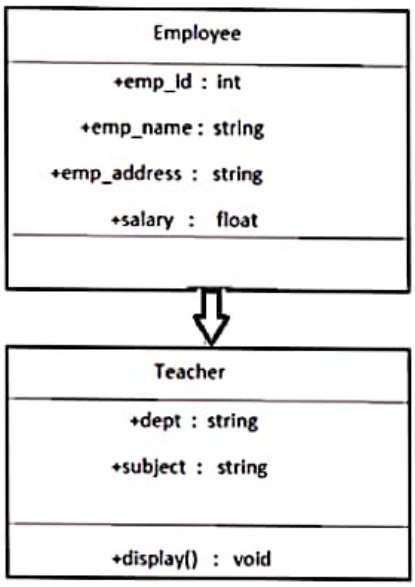
PROGRAM NO – 34

AIM: To develop a Java program to create a class 'employee' with data members empid, name, salary,
address and constructors to initialize the data members. Create another class 'teacher' that inherit the properties of class employee and contain its own data members department, subjects taught and constructors to initialize these data members and also include display function to display all the data members. Use array of objects to display details of N teachers.



PROGRAM

```
import java.util.Scanner;
class Employee                    //Parent class
{
   int empid;
   String name,address,subject;
   double salary;

   Employee(int no, String na, double sal, String addr)  // Parent class constructor
   {
      empid = no;
      name = na;
      salary = sal;
      address = addr;
```

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```
    }
}
class Teacher extends Employee          //Child class
{
    String dept,sub;
    Teacher(int no, String na, double sal, String addr, String dep,String sub)   //Child class
constructor
    {
        super(no,na,sal,addr);    // Passing data to parent class constructor
        dept= dep;
        subject= sub;
    }
    void display()
    {
        System.out.println("Employee id: "+empid);
        System.out.println("Name: "+name);
        System.out.println("Salary: "+salary);
        System.out.println("Address: "+address);
        System.out.println("Department: "+dept);
        System.out.println("Subject: "+subject);
    }
}
class EmpInheritance
{
    public static void main(String[] args)
    {
        System.out.println("\nEnter the no. of Employees");
        Scanner s = new Scanner(System.in);
        int num = s.nextInt();
        Teacher arr[]=new Teacher[num];
        for(int i =0;i<num;i++)
        {
            System.out.println("\nEnter Employee-"+(i+1)+" details: id,Employee Name, Salary, Address,
```

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```
                                Department,Subject");
        int empid=s.nextInt();
        String name=s.next();
        double salary=s.nextDouble();
        String address=s.next();
        String dept=s.next();
        String subject=s.next();
        arr[i]=new Teacher(empid,name,salary,address,dept,subject); // Passes values to child class
constructor
    }
    System.out.println("\n**Informations of all the employee's**");
    for(int i=0;i<num;i++)
    {
        System.out.println("\nEmployee-"+(i+1)+" details");
        System.out.println("----------------------");
        arr[i].display();
    }
    s.close();
  }
}
```

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NOTES

super keyword

- When a superclass has its fields declared as private, these fields are private to the superclass and are not accessible by its subclasses using the conventional style.
- Whenever a subclass needs to refer to its immediate superclass, it can do so by using the keyword **super**. We use super in two ways:
  • To call superclass constructor: A subclass can call a constructor defined by its superclass as
  follows:

```
super(arg-list );
```

- Here, arg-list specifies arguments, if any, needed by the constructor of its superclass.
- super() must be the 1st statement inside a subclass's constructor.
- Eg: consider the program below.

```
class Animal
{
        Animal ( )
        {
                System . out . p r i n t l n ( " Animal  i s  c r e a t e d " ) ;
        }
}
class Dog extends Animal
{
        Dog ( )
        {
                super ( ) ;
                System . out . p r i n t l n ( "dog  i s  c r e a t e d " ) ;
        }
}
class TestSuper1
{
        public static void main ( String args [ ] )
        {
                Dog d=new Dog ( ) ;

        }
}
```

Output

animal is created

dog is created

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PROGRAM NO – 35

AIM: To develop a Java program to demonstrate multilevel Inheritance. Create a class "Principal"(name) which is inherited by the class "Teacher" (tchrId) which in turn is inherited by class "Student" (rno,display())

PROGRAM

```
class Principal
{
   String name="Rakesh";
}
class Teacher extends Principal
{
   int tchrId=56;
}
class Student extends Teacher
{
   int rno=78;
   void display()
   {
      System.out.println("Principal name :"+name);
      System.out.println("Teacher-id :"+tchrId);
      System.out.println("Student-roll no :"+rno);
   }
}
class Multilevel
{
   public static void main(String args[])
   {
      Student ob=new Student();
      ob.display();
   }
}
```

OUTPUT

```
javac Multilevel.java
PS E:\PT-Java> java Multilevel

Principal name :Rakesh
Teacher-id :56
Student-roll no :78
```

NOTES

Multilevel Inheritance

When there is a chain of inheritance, it is known as multilevel inheritance.
"We create objects for the class at the lowest level"

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In the example given below, Animal class is inherited by Dog class and Dog class is inherited by BabyDog class.

```
class Animal
{
        void eat( )
        {
                System.out.p r i n t l n ( " e a t ing . . . " ) ;
        }
}
class Dog extends Animal
{
        void bark( )
        {
                System.out.p r i n t l n ( " barking . . . " ) ;
        }
}
class BabyDog extends Dog
{
        void weep( )
        {
                System.out.p r i n t l n ( "weeping . . . " ) ;
        }
}
class TestInheritance
{
        public static void main ( St ring args [ ] )
        {
                BabyDog d = new BabyDog ( ) ;
                d.weep ( ) ;
                d.bark ( ) ;
                d.eat( ) ;
        }
}
```

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Leave space as was needed for Pgm 35

PROGRAM NO – 36

AIM: To develop a Java program to demonstrate multilevel Inheritance. Create a class "Manager"(salary) which is inherited by the class "TeamLead" (teamName) which in turn is inherited by class "Developer" (prjcId,putdata())

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PROGRAM NO – 37

AIM: To develop a Java program to demonstrate use of "super" keyword to access super class constructor. Create a class "SuperParent" (rollno) which is inherited by the class "SuperChild"(name,display()).

PROGRAM

```
class SuperParent
{
   int rollno;
   SuperParent(int rno)
   {
      rollno=rno;
   }
}
class SuperChild extends SuperParent
{
   String name;
   SuperChild(int rno, String name1)
   {
      super(rno); //Calling super class constructor
      name=name1;
   }
   void display()
   {
      System.out.println("Roll no.:"+rollno);
      System.out.println("Name.:"+name);
   }
}
class SuperDemo
{
   public static void main(String args[])
   {
      SuperChild ob=new SuperChild(23,"Deepthy");
      ob.display();
   }
}
```

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Leave space as was needed for Pgm -37

PROGRAM NO – 38

AIM: To develop a Java program to demonstrate use of "super" keyword to access super class constructor. Create a class "Parent" (pid) which is inherited by the class "Child"(course,putdata()).

PROGRAM

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